

Nishal Chandra Reddy

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5 years building and owning production systems end to end. Java, Spring Boot and Node.js microservices at scale. Kafka event pipelines and LLM extraction pipelines running on live traffic. Have led multi-service initiatives spanning six services and mentored engineers through architecture reviews. Comfortable at every layer: backend architecture, cloud infrastructure and IaC, React front ends. M.S. in Computer Science, Arizona State University (2023–2025).

TECHNICAL SKILLS

Programming Languages: Java 8/11/17, JavaScript, Python, TypeScript, Go, C/C++, PHP, SQL, Bash

Backend & APIs: Spring Boot (Spring Security, Spring Data JPA, Hibernate, Spring Cloud, Actuator), Node.js, Express, FastAPI, Laravel, REST, GraphQL, gRPC, WebSockets, OpenAPI, OAuth 2.0, JWT, Keycloak

Build & Testing: Maven, JUnit, Mockito, Jest, Vitest, pytest, TDD, BDD, SonarQube

Frontend: React, Next.js, Vite, Tailwind CSS, Material UI

AI / ML: LLM integration (Anthropic Claude, AWS Bedrock, Google Gemini), RAG & vector search, prompt engineering, structured extraction, evaluation & A/B testing, PyTorch, scikit-learn

Data & Messaging: Kafka, SQS/SNS, PostgreSQL, MySQL, MongoDB, Cassandra, Redis, Elasticsearch, Neo4j

Cloud & DevOps: AWS (ECS/Fargate, Lambda, EC2, S3, SES, Secrets Manager, CloudWatch, Textract), GCP, Docker, Kubernetes, Terraform, Ansible, GitHub Actions, GitLab CI, Jenkins

Observability: Prometheus, Grafana, ELK, CloudWatch alarms

EDUCATION

Master of Science, Computer Science | Arizona State University, Tempe, AZ | GPA 3.77 May 2025

Distributed Database Systems · Cloud Computing · Statistical Machine Learning · Data Mining

Bachelor of Engineering, Computer Science | Bangalore Institute of Technology, Bengaluru, India May 2019

PROFESSIONAL EXPERIENCE

Software Developer | Ecohome.one — Remote, US Aug 2025 – Present

- Led the **GDPR erasure program across six services**, designed the specs and built a coordinator that fans a single deletion request out to idempotent erasure workers in each service, using a MySQL state machine with per-household locks and single-active-request enforcement.
- Build and operate the AI receipt-extraction platform, a Node.js service on **ECS Fargate** that turns unstructured purchase emails into structured inventory records through a Gemini pipeline with vision OCR for PDF and image attachments.
- Hardened the pipeline against LLM failure with a **four-tier fallback chain**, per-stage heuristic extractors, a config-swappable Bedrock/Claude path, SQS re-queue and auto-requeue on provider recovery.
- Built two production services solo, an **Alexa integration service** with request-signature verification, OAuth account linking and circuit breakers, shipped with its own Terraform IaC and CI/CD and **EcohomeWeb**, a React 18 and Vite SPA. Also built the eBay resale marketplace in the core Express API, with a listing state machine, scheduled delist workers and HMAC webhook verification.

Senior Software Developer | Comviva — Bengaluru, India May 2021 – Jul 2023

- Drove the architectural redesign of the **MobiLytx Loyalty & Rewards** platform, decomposing a monolith into modular **Spring Boot** services (authentication, multi-channel notifications, rewards) along Domain-Driven Design bounded contexts and rebuilding the customer-facing campaign console as a component-driven React SPA. Increased system throughput 60% and accelerated feature delivery 35%.
- Implemented the service layer on **Spring Data JPA and Hibernate** over PostgreSQL, with **Spring Cloud** for centralized config and service discovery, Actuator metrics scraped by Prometheus and Spring Security with Keycloak (OAuth 2.0 / JWT) enforcing role-based access across tenants.
- Architected event-driven data pipelines on **Kafka** for asynchronous inter-service communication, reliably processing **700K+ daily transactions at sub-200 ms latency**.
- Migrated the platform from on-premises infrastructure to AWS as a SaaS deployment, containerizing services with **Docker and Kubernetes** and introducing centralized logging via the ELK stack, reducing infrastructure cost 40%.
- Led a **cross-functional team of five engineers**, running architecture reviews, mentoring junior developers and aligning delivery with product goals; standardized Maven multi-module builds and JUnit/Mockito coverage gates in CI.

Software Developer | Comviva — Bengaluru, India Aug 2019 – Apr 2021

- Built the **Toll-Free Data Access** system, network traffic monitoring in C/C++, Redis-backed session management and reverse-proxy routing logic in Node.js, serving zero-rated data access at carrier scale.
- Shipped rule evaluation, eligibility logic and reward fulfillment paths into the **Java and Spring Boot campaign rules engine** driving personalized reward programs for carrier customers.
- Automated CI/CD with GitLab CI, Jenkins and SonarQube, adding API contract tests and BDD to the pipeline; instrumented services with Prometheus and Grafana. Deployment frequency rose 40% and production errors fell 30%.

SELECTED PROJECTS

Swarm Intelligence-Based Distributed Database — *Kafka, MongoDB, Docker Swarm, Prometheus* 2025

- Built a fault-tolerant distributed database ingesting **10M+ IoT data points daily** via Kafka and MongoDB sharding, sustaining 99.99% availability and absorbing 5x traffic spikes through automated node scaling.
- Improved query performance up to 45% by applying **Ant Colony Optimization** to dynamic query routing and Particle Swarm Optimization to adaptive resource allocation.

NetGuardian AI — Predicting Problematic Internet Use — *Autoencoders, scikit-learn, Ensemble Models* 2024

- Built an end-to-end ML pipeline predicting severity of problematic internet use from **10,000+ hours** of high-dimensional actigraphy time-series, using autoencoders and statistical summarization for feature extraction.
- Raised model accuracy 20% with boosting and bagging ensembles, and tuned Quadratic Weighted Kappa scoring with stratified cross-validation for stable performance across user groups.